

Tello Drone Control and Analysis using Python, Netcat, and Wireshark

Introduction

My name is Abhilash. In this project, I performed a practical experiment on the Ryze Tello drone to understand how IoT devices communicate over a network. The goal was to control the drone using different methods and analyze the communication using cybersecurity tools.

This experiment demonstrates UDP communication, network analysis, and potential security risks in IoT devices.



1. Mobile Control (Initial Setup)

- First, I connected my mobile phone to the drone's Wi-Fi network (TELLO-XXXX). The drone acts as a wireless access point.
- Using the official Tello mobile application, I made the drone take off and hover for a few seconds.
- This shows that the drone accepts commands from the app using UDP communication.
- After this step, I disconnected the phone to allow the laptop to take control.

2. Network Verification

- After connecting my laptop to the drone Wi-Fi, I verified connectivity using:
- ping 192.168.10.1
- This command sends ICMP packets to the drone. A successful response confirms that the system can communicate with the drone.
- The TTL value of 255 indicates that the device is on the same local network and is an embedded system.

3. Kali Linux Virtual Machine Issue

- I attempted to use Kali Linux inside a virtual machine (UTM). However, when I checked the IP address using:

`ip a`

The system showed an IP in the range 192.168.64.x.

This means the VM is using a NAT network and cannot directly access the drone network (192.168.10.x).

This demonstrates an important concept: virtualization can isolate network interfaces, preventing direct communication with external devices.

4. Python-Based Drone Control

- I used Python to send UDP commands directly to the drone.

Commands used:

1. python3

Then in Python:

2. `import socket` //Importing the socket module.

```

3. tello = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
   //Creating a UDP socket.
4. tello.bind(('', 9000)) //Binding to port 9000 to receive
   responses.
5. tello.sendto(b'command', ('192.168.10.1', 8889))
   print(tello.recv(1024)). //Enabling SDK mode on the drone.
6. tello.sendto(b'takeoff', ('192.168.10.1', 8889))
   print(tello.recv(1024)) //Sending takeoff command.
7. tello.sendto(b'land', ('192.168.10.1', 8889))
   print(tello.recv(1024)) //Landing the drone safely.

```

Explanation:

- socket.AF_INET: IPv4 communication
- SOCK_DGRAM: UDP protocol
- bind(): receive responses
- command: enables SDK mode
- takeoff/land: control commands

This demonstrates direct low-level control using UDP.

5. Netcat Method (Direct Packet Injection)

I also used Netcat to send commands directly from the terminal:

- ```

1. echo "command" | nc -u 192.168.10.1 8889
2. echo "takeoff" | nc -u 192.168.10.1 8889
3. echo "land" | nc -u 192.168.10.1 8889

```
- This method does not require programming.
  - Here I am using Netcat to directly send UDP packets to the drone. This method does not require any programming and demonstrates low-level packet injection.
  - It shows that anyone on the same network can send commands, which is a security risk.
  - This also highlights a security concern, as any device connected to the network can send control commands without authentication.

## 6. Wireshark Analysis

To analyze network traffic, I used Wireshark:

**wireshark**

Then applied filter:

**udp.port == 8889**

- While controlling the drone, I observed packets containing commands like **"takeoff"** and **"land"**.
- This shows that the communication is not encrypted and can be intercepted.
- This reveals a potential vulnerability, where an attacker could intercept or replicate commands.

## 7. Conclusion

This experiment demonstrates how IoT devices like drones use UDP for communication. While UDP allows fast control, it lacks authentication and session management.

This creates potential security risks, as unauthorized users can send commands or intercept traffic.

Overall, this project helped me understand networking, cybersecurity concepts, and real-world device communication.